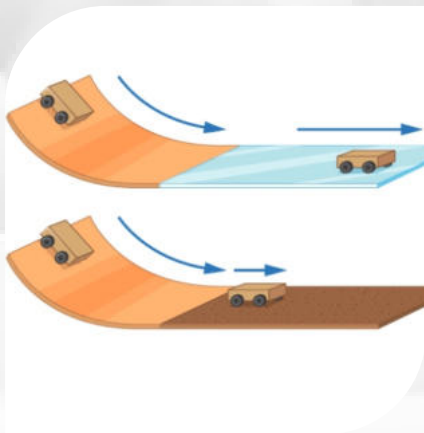
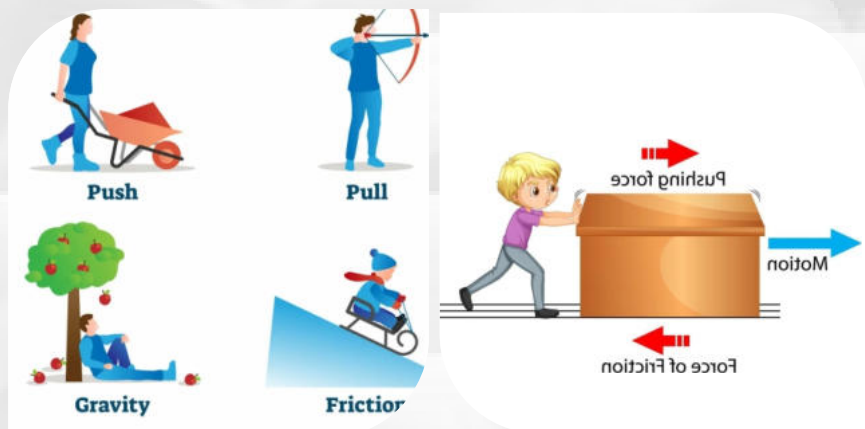


UNIT 05

FORCES AND MATTER



FORCE

“Force is a push or pull that changes or tends to change the state of rest or uniform motion of an object or changes the direction or shape of an object”.

Examples

- ▶ Force is needed to move a car.
- ▶ Force causes the spring to stretch.
- ▶ We need force to move some luggage.
- ▶ If you bend your plastic ruler, you will change its shape.

**EXPLANATION**

Force is a basic concept in physics that describes an interaction between two objects that causes a change in the motion of one or both of the objects. In physics, forces are often used to explain why they do objects move. For example, if you push a book across a table, the force of your hand on the book is what causes it to move. Forces can also cause objects to change direction, speed up, slow down, or remain at rest.

CLASSIFICATION OF FORCE

Forces can be classified into two main categories:

Contact forces

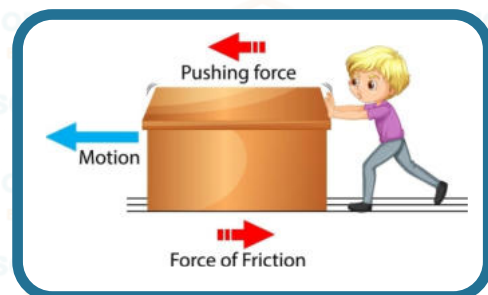
Non-contact forces.

CONTACT FORCES

Contact forces can be further classified into four types: friction, tension, normal force, and air resistance.

FRICTION

Friction is a force that opposes motion between two objects that are in contact. Friction is generated by the microscopic roughness of the surfaces in contact. The force of friction depends on the type of surface and the force that is applied. The coefficient of friction is a measure of the strength of the frictional force.

**TENSION FORCE**

Tension is a force that is transmitted through a rope, cable, or string. Tension is the force that is required to keep the rope taut. The force of tension is equal to the force applied to one end of the rope, and it decreases as the rope is stretched.

NORMAL FORCE

The normal force is a force that acts perpendicular to a surface and is exerted by the surface on an object that is in contact with it. The normal force is equal in magnitude to the weight of the object, but in the opposite direction. The normal force is essential in determining the stability of an object and the direction of motion.

AIR RESISTANCE

Air resistance is a force that opposes motion through the air. The force of air resistance depends on the velocity of the object, the surface area, and the density of the air. Air resistance can be reduced by decreasing the surface area or by increasing the speed of the object.

NON-CONTACT FORCES

Air resistance is a force that opposes motion through the air. The force of air resistance depends on the velocity of the object, the surface area, and the density of the air. Air resistance can be reduced by decreasing the surface area or by increasing the speed of the object.

GRAVITATIONAL FORCE

Gravitational force is a force that acts between two objects that have mass. The force of gravity depends on the masses of the objects and the distance between them. Gravitational force is the force that keeps the planets in orbit around the sun.

ELECTROMAGNETIC FORCE

Electromagnetic force is a force that acts between charged objects. The force of electromagnetic force depends on the charge and the distance between the objects. Electromagnetic force is responsible for the forces between charged particles, such as ions and electrons.

NUCLEAR FORCE

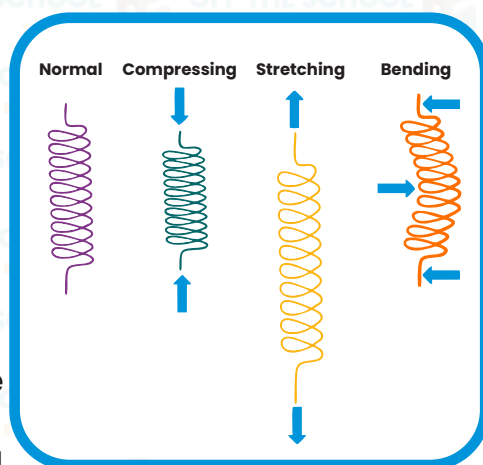
Nuclear force is a force that acts between the nuclei of atoms. The force of nuclear force depends on the distance between the nuclei and the charge of the nuclei. Nuclear force is responsible for the stability of the nucleus and the binding of the protons and neutrons.

FORCES ACTING ON SOLIDS

Solids have definite shapes and sizes; however, it is possible to change their shapes and sizes by applying external forces. When the external force is removed, the object tends to return to its original shape and size. This behavior is called elastic behavior. Solids can be stretched, squashed, bent or twisted.

STRETCHING SPRINGS

Springs are often used to model elastic materials, which are able to return to their original shape after being deformed by an applied force. As springs are designed to stretch a long way when force is applied, therefore it is easy to measure changes in their lengths. Consider a spring hung from a rigid support, so that its top end is fixed. Weights are hung on the other end of the spring. These are called loads. As load is increased, the spring is stretched and its length increases. When the load is removed, the spring returns to its original length. This is called elastic change. The stretching of springs has many practical applications, such as in mechanical systems, suspension systems in vehicles, and in measuring instruments, such as spring scales. By understanding the behavior of springs when they are stretched, we can design and engineer systems that make use of this relationship in a controlled and predictable manner.



EXTENSION OF SPRING

The length of spring increases as the force (load) increases. This increase in length of spring is known as extension.

Hence;

$$\text{Length of stretched spring} = \text{Original length} + \text{Extension}$$

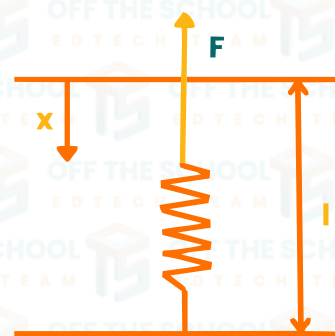
ELASTICITY

Elasticity is a property of materials that refers to their ability to return to their original shape after being deformed by an applied force. In physics, elasticity is a fundamental concept that is used to describe the behavior of materials under stress and strain. Imagine a rubber band, which can be stretched and pulled but will eventually return to its original shape when the force is removed. This is an example of elastic behavior. The amount by which a material can be stretched or compressed before it breaks is known as its elastic limit.

HOOKE'S LAW

INTRODUCTION

Robert Hooke, a scientist and inventor, was interested in springs for two reasons: Spring is useful in making balances. Hooke wanted to make a very sensitive and accurate weight machine or balance. He also realized that a spiral spring could be used to control a clock or wrist watch.



STATEMENT

“Within elastic limit, the displacement produced in the spring is directly proportional to the applied force”.

Mathematically

if “**F**” is the applied force and ‘**x**’ is the displacement (extension) in the spring then the equation for Hooke's law may be written as:

$$F \propto x$$

$$F = - kx$$

Where **k** is spring constant (stiffness of spring).

APPLICATION OF HOOKE'S LAW

Hooke's law is applicable to all kinds of deformation and all types of matter i.e., solids, liquids or gases within a certain limit. This limit tells the maximum force or stress that can be safely applied on a body without causing permanent deformation in its length, volume or shape. In other words, it is a limit within which a body recovers its original length, volume or shape after the deforming force is removed. It is important to note that Hooke's law only applies to springs and other elastic materials within their elastic limit, meaning that they will return to their original shape when the force is removed. If the force exceeds the elastic limit, the material will not return to its original shape and may break.

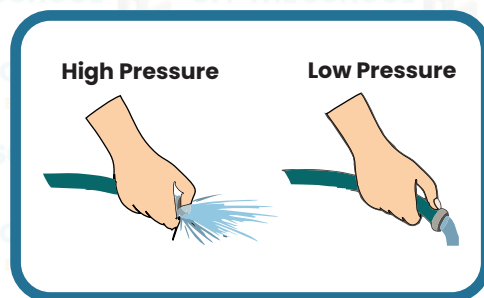
PRESSURE

DEFINITION

THE FORCE ACTING NORMALLY PER UNIT AREA ON THE SURFACE OF A BODY IS CALLED PRESSURE.

FORMULA:

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}}$$



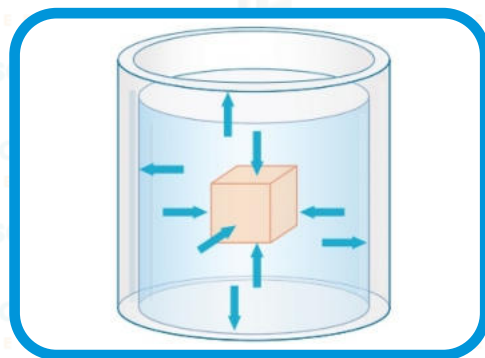
HOW TO CALCULATE PRESSURE?

If F is the magnitude of a force exerted perpendicular to a given surface of area A , then the pressure P equals to the force divided by the area. As the force is measured in Newton's (**N**) and area in square meters (**m²**). Therefore, in the SI system, the unit of pressure is Newton per square meter (**N/m²**). It is also known as Pascal (**PA**):

$$1 \text{ Pa} = 1 \text{ Nm}^{-2}$$

PRESSURE IN FLUIDS

A fluid is a collection of molecules that are randomly arranged and held together by weak cohesive forces and by forces exerted by the walls of a container. Both liquids and gases are fluids. The pressure exerted by fluids is known as fluid pressure. It acts in all directions. This is because the molecules of fluids move around in all directions; causing pressure on every surface they collide with.



SWIMMER FEELS FLUID PRESSURE IN SWIMMING POOL

The fluid pressure felt by a swimmer in a swimming pool is caused by the weight of the water above them. The deeper a swimmer goes into the pool, the greater the fluid pressure they experience. This pressure increases with the depth and density of the fluid, as well as the gravity. The pressure at the bottom of a pool is greater than at the surface, and this difference in pressure creates a force that pushes the swimmer upward. The swimmer feels this force as an increase in lightness, which helps them float.

FACTORS AFFECTING ON PRESSURE

Depth: The deeper the fluid, the greater the pressure. This is because there is more fluid above and the weight of this fluid increases with depth.

Density: The denser the fluid, the greater the pressure. For example, water is denser than air and will create greater pressure.

Gravity: The strength of the gravitational force affects the pressure of the fluid. The pressure is greater in regions where gravity is stronger, such as at sea level compared to on a mountaintop.

Temperature: The temperature of a fluid can affect its pressure. For example, when a fluid is heated, its volume increases, leading to a decrease in pressure.

Effects of Velocity on Pressure: Pressure changes due to changes in velocity, such as waves, can also affect the pressure of a fluid.

Pressure = depth × density × acceleration due to gravity

$$P = \rho gh$$

HYDRAULIC MACHINE

A hydraulic machine is a device that uses a liquid, usually oil, to transmit power from one place to another. Hydraulic machines are commonly used in construction equipment, industrial machines, and automotive systems, among other applications. The benefits of hydraulic machines include their ability to generate large forces with relatively small inputs, their versatility and adaptability to different tasks, and their durability and reliability.



PASCAL'S LAW

It's named after **Blaise Pascal**, a French mathematician and physicist who first stated this law in 1647.

STATEMENT

"The pressure applied to an enclosed fluid is transmitted equally in all directions and acts with equal force on equal areas."

In simpler terms, this law means that if you press down on one point in a container filled with fluid, the pressure will be distributed evenly throughout the entire fluid. This law has important practical applications, including in hydraulic systems, where it is used to design machines that use fluid pressure to generate force and motion, such as car lifts and construction equipment.

EXPLANATION

Consider a water filled in a glass vessel having holes around its surface. When you apply force through the piston the water rushes out of the holes with the same pressure. The force applied on the piston exerts pressure on water. This pressure is transmitted equally throughout the liquid in all directions. In general, this law holds good for fluids and gases.

EXAMPLE

A hydraulic machine works on this principle. Hydraulic brakes, car lifts, hydraulic jacks, forklifts, and other machines make use of this principle.

HYDRAULIC PRESS

A hydraulic press is made of two pistons connected by a liquid-filled pipe as demonstrated by Pascal's law. A force of magnitude **F₁** is applied to a small piston of surface area **A₁**. The pressure is transmitted through an incompressible liquid to a larger piston of surface area **A₂**. Because the pressure must be the same on both sides.



$$P = P_1 = P_2$$

As we know

$$P = \frac{F}{A}$$

So,

$$\frac{F}{A} = \frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$$P = \frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$$F_2 = \frac{A_2}{A_1} \times F_1$$

By designing a hydraulic press with appropriate areas A_1 and A_2 , a large output force can be applied by means of a small input force. Each side of this equation is the work done by the force. Thus, the work done by F_1 on the input piston is equal to the work done by F_2 on the output piston. Thus, the principle of conservation of energy applies in the hydraulic press

